Pavlo Radiuk

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I work on explainable artificial intelligence: methods that map what a deep network has learned onto the features clinicians and engineers already use, so that its decisions can be checked and audited. I apply them in medical imaging, UAV inspection of energy infrastructure and urban traffic control, and I am a Researcher in the Horizon Europe project U_CAN.

Positions

2025 – present **Associate Professor**, Khmelnytskyi National University
2024 – present **Researcher, Horizon Europe project U_CAN**, Khmelnytskyi National University
2022 – 2025 **Senior Lecturer**, Khmelnytskyi National University
2022 – present **Data Scientist (advisory)**, Horizon Dynamics

Education

2025 – 2027 **D.Sc. (habilitation level)**, Information Technologies, Khmelnytskyi National University, in progress
2022 **Ph.D.**, Computer Science, Khmelnytskyi National University
2017 **M.Sc.**, Mathematical and Computer Modelling, Khmelnytskyi National University, with distinction

Research projects

2024 – 2028 **Ukraine towards Carbon Neutrality (U_CAN)**, Researcher, Khmelnytskyi traffic pilot
Horizon Europe, EU Mission for Climate-Neutral and Smart Cities, grant agreement 101148374



Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Climate, Infrastructure and Environment Executive Agency (CINEA). Neither the European Union nor the granting authority can be held responsible for them.

2024 – 2026 **Intelligent System for Recognising Defects of Green-Energy Objects Using UAVs**, Researcher (executor)
Ministry of Education and Science of Ukraine, No. 0124U004665

Selected publications

76 works: 38 journal articles, 34 conference papers, 2 book chapters and 2 preprints. Google Scholar h-index 17 (1,158 citations) · Scopus h-index 10 · Web of Science h-index 7 (as of Oct 2026). Own name in bold; † marks a mentored junior researcher.

2026 **CARDIA-X: Global Semantic Transition and Rough-Set Rules for Auditable Post Hoc Electrocardiographic Explainability**
P. Radiuk, O. Barmak, L. Klymenko, I. Krak. *AI* 7(9):376. doi:10.3390/ai7090376

2026 **Equivariant Transition Matrices for Explainable Deep Learning: A Lie Group Linearization Approach**
P. Radiuk, O. Barmak, L. Bedratyuk, I. Krak. *Machine Learning and Knowledge Extraction* 8(4):92. doi:10.3390/make8040092

2025 **Criticality Assessment of Wind Turbine Defects via Multispectral UAV Fusion and Fuzzy Logic**
P. Radiuk, B. Rusyn, O. Melnychenko, T. Perzynski, A. Sachenko, S. Svystun†, O. Savenko. *Energies* 18(17):4523. doi:10.3390/en18174523

2025 **Hierarchical Deep Learning Model for Identifying Similar Targets in UAV Imagery**
D. Borovyk†, O. Barmak, **P. Radiuk**, I. Krak. *Drones* 9(11):743. doi:10.3390/drones9110743

2024 **Intelligent Integrated System for Fruit Detection Using Multi-UAV Imaging and Deep Learning**
O. Melnychenko, L. Scislo, O. Savenko, A. Sachenko, **P. Radiuk**. *Sensors* 24(6):1913. doi:10.3390/s24061913

2017 **Impact of Training Set Batch Size on the Performance of Convolutional Neural Networks for Diverse Datasets**
P. Radiuk. *Information Technology and Management Science* 20(1). doi:10.1515/itms-2017-0003

Service and recognition

PC member DEXA 2024–2026 · WISE 2024, 2026 · MobiWis 2026
Editorial Guest Editor, Sustainability (MDPI), 2024–2026; Volume Editor, CEUR Workshop Proceedings, Vol. 4141 (ExplAI-2025), 2025; Editorial Board Member and Handling Editor, Journal of Intelligent & Fuzzy Systems, 2023–2024.
Organising Organising Committee member, ExplAI workshop, 2025–2026; Lead organiser, U_CAN international conference on climate-neutral cities, 2026.
Peer review 450+ reviews for journals and conferences.
Awards Scholarship of the Cabinet of Ministers of Ukraine for Young Scientists, 2022–2024; Best Paper Award for Ukrainian authors, IDAACS-2023, 2023.
Teaching 8 current courses at bachelor's, master's and Ph.D. level, one taught in English.

Collaboration

Partners Open to Horizon Europe, MSCA and EuroHPC consortia as XAI methods lead, Medical AI partner, Energy & UAV inspection and Ukrainian pilot site (PIC 964860456). Partner kit: radiukpavlo.github.io/partner-kit.